

QC BOOT CAMP MINI-PROJECTS, FIRST BATCH

Instructions: Pick and solve/implement one project either from the list below, or a project that you discussed with me (and I approved).¹ Create a [github](#) repo from your submission and send it to me by the **end of Monday, June 22**.

Alternative/Advanced topics: If you already have a project idea, feel free to discuss it with me, and if I approve it, then you can do that project instead.

I am also happy to recommend more advanced topics too, if you feel like challenging yourself. Such topics are: [decoded quantum interferometry](#), [linear algebra on quantum computers](#), and [quantum walk search/Markov chain based quantum algorithms](#), to name a few, but I can also help you look for a topic that interests you the most.

MINI-PROJECT #1: $n^2 \times n^2$ SUDOKU SOLVER WITH GROVER'S ALGORITHM

Write a Qiskit function that takes an $n^2 \times n^2$ instance of a [Sudoku puzzle](#) and outputs a solution if there is one, with high probability, using Grover's algorithm.

In order to complete this task, it is enough to make it work for $n = 2$. However, I encourage you to try to implement it for larger values of n .

You are allowed to use multi-controlled Z gates in the diffuser only. Otherwise, the construction may use any number of ancillas, arbitrary 1-qubit gates, CX , and Toffoli gates. No classical bit and measurements allowed.

Expectations:

- Full github repo: README and other relevant documentation.
- Working Qiskit code with benchmarking for $n = 2$.

MINI-PROJECT #2: SUBSET-SUM PROBLEM WITH GROVER'S ALGORITHM

Write a function, that takes two inputs, a finite set of positive integers, $\{x_0, x_1, \dots, x_{n-1}\}$, and another positive integer t , and outputs a set $I \subseteq \{0, 1, \dots, n-1\}$, such that $\sum_{i \in I} x_i = t$, or decides, with high probability, that no such set exists.

In order to complete this task, it is enough to make it work for $n \leq 4$ and $x_i < 8$. However, I encourage you to try to implement it for all inputs.

You are allowed to use controlled phase gates. Otherwise, the construction may use any number of ancillas, arbitrary 1-qubit gates and Toffoli gates, if needed. No classical bit and measurements allowed.

Expectations:

- Full github repo: README and other relevant documentation.

Date: June 2, 2026.

¹You only need to submit one, but I'm happy to give feedback if you do more of them.

- Working Qiskit code with benchmarking for $n \leq 4$ and $x_i < 8$.

MINI-PROJECT #3: MULTI-CONTROLLED U GATE

Write a Qiskit function that takes two inputs: a positive integer n and a matrix $U \in U(2)$ and outputs a quantum circuit on $n + 1$ qubits, possibly with further ancillas, that implements a multi-controlled U gate, $C^n U$, that is

$$C^n U |x\rangle_n |y\rangle_1 = \begin{cases} |x\rangle_n U |y\rangle_1, & \text{if } x = (1, 1, \dots, 1), \\ |x\rangle_n |y\rangle_1, & \text{otherwise.} \end{cases}$$

The construction may use any number of ancillas, arbitrary 1-qubit gates, CX , and Toffoli gates. No classical bit and measurements allowed.

Hint: There are a lot of papers written on the subject, so start with a literature search (and/or asking me ☺).

Expectations:

- Full github repo: README and other relevant documentation.
- Working Qiskit code with examples for small values of n .
- Analysis of the various complexities and resources used (at least for gate count, gate depth, and number of ancillas), in the “big- O ” sense, as $n \rightarrow \infty$.

MINI-PROJECT #4: QUANTUM CALCULATOR

Below we use the numerical labeling on the d -qubit computational basis, that is

$$|x_0\rangle |x_1\rangle \cdots |x_{d-1}\rangle =: |x\rangle_d,$$

if $\overline{x_0 x_1 \cdots x_{d-1}}$, is the binary representation of x .

Write a Qiskit function that takes one input: a positive integer d and outputs a quantum circuit, QCalc, on $3d + 1$ qubits, possibly with further ancillas, that implements the following:

$$\text{QCalc} |x\rangle_d |y\rangle_d |z\rangle_1 |0\rangle_d = \begin{cases} |x\rangle_d |y\rangle_d |z\rangle_1 |x + y \bmod 2^d\rangle_d, & \text{if } z = 0, \\ |x\rangle_d |y\rangle_d |z\rangle_1 |x \cdot y \bmod 2^d\rangle_d, & \text{if } z = 1 \end{cases}$$

The construction may use any number of ancillas, arbitrary 1-qubit gates, CX , and Toffoli gates. No classical bit and measurements allowed.

Hint: You can use Draper’s Quantum Fourier Transform based adder; see [Draper: Addition on a Quantum Computer](#).

Expectations:

- Full github repo: README and other relevant documentation.
- Working Qiskit code with examples for small values of d .
- Analysis of the various complexities and resources used (at least for gate count, gate depth, and number of ancillas), in the “big- O ” sense, as $d \rightarrow \infty$.

MINI-PROJECT #5: EIGENVECTOR PREPARATION

Write a Qiskit function that takes the following inputs:

- (1) An n -qubit diagonal unitary, that is, $U|x\rangle_n = e^{2\pi i\theta(x)}|x\rangle_n$, such that all values of $e^{2\pi i\theta(x)}$ are distinct. You can
- (2) A number $d \in \mathbb{Z}_+$, with the promise that $2^d\theta(x) \in \mathbb{Z}$.
- (3) A value $t \in [0, 1]$, with the promise that there exists one (and thus, only one) $x \in \mathbb{F}_2^n$, such that $e^{2\pi i\theta(x)} = e^{2\pi it}$.

The output of the function is a quantum circuit that prepares ("finds") $|x\rangle$, up to an error of $O(\frac{1}{2^n})$.

In order to complete this task, it is enough to make it work for $n, d \leq 3$. However, I encourage you to try to implement it for larger values of n, d .

Hint: Use Quantum Phase Estimation to mark $|x\rangle$ and then use it as the marker oracle in Grover's algorithm.

You can input U as 2^n -dimensional vector of (ordered) eigenvalues and then you can use the built-in function [qiskit.circuit.library.Diagonal](#) and [ControlledGate](#). You can also do some classical preprocessing on U , such as computing its powers. No direct ("classical") search allowed though, of course. The construction may also use any number of ancillas, 1-qubit gates, multi-controlled phase (*MCP*) and *X* (*MCX*) gates with any number of controls, and the built-in version of the Quantum Fourier Transform (and its inverse). If you feel like you want to use another built-in functionality in Qiskit, then please reach out to me and I let you know how I feel about it. ☺ No classical bits and measurements allowed.

Expectations:

- Full github repo: README and other relevant documentation.
- Working Qiskit code with examples for $n, d \leq 3$.

MINI-PROJECT #6: ORACLE FOR SHOR'S ALGORITHM

Write a Qiskit function that takes two inputs, $a, N \in \mathbb{Z}_+$, such that $a < N$ and outputs the oracle

$$U|x\rangle_1|y\rangle_n = \begin{cases} |x\rangle_1|ay \bmod N\rangle_n, & \text{if } x = 1 \text{ and } y < N, \\ |x\rangle_1|y\rangle_n, & \text{otherwise.} \end{cases}$$

where $n = \lceil \log_2(N) \rceil$. **It is enough to make it work for $N = 15$.** However, I encourage you to try to implement it for arbitrary N .

The construction may only use any number of ancillas, 1-qubit gates, multi-controlled phase (*MCP*) and *X* (*MCX*) gates with any number of controls, and the built-in version of the Quantum Fourier Transform (and its inverse). No classical bit and measurements allowed.

Hint: There are a lot of papers written on the subject, so start with a literature search (and/or asking me ☺). For example, this is an old (and no longer SOTA), but good implementation that you can use: [Circuit for Shor's algorithm using \$2n + 3\$ qubits](#).

Expectations:

- Full github repo: README and other relevant documentation.
- Working Qiskit code with a demo for $N = 15$.

MINI-PROJECT #7: QUANTUM STATE PREPARATION

Write a Qiskit function that takes a 2^n dimensional vector, $\phi \in \mathbb{C}^{2^n}$, such that $\|\psi\|_2 = 1$, and outputs a circuit, U , such that

$$U|0\rangle_n = \sum_{x=0}^{2^n-1} \psi_x |x\rangle_n.$$

The construction may use any number of ancillas, arbitrary 1-qubit gates and multi-controlled R_Z gates, (the number of controlled may be arbitrarily large). No classical bit and measurements allowed.

Expectations:

- Documentation.
- Working Qiskit code for all n and demo for $n = 3$.

MINI-PROJECT #8: QUANTUM ROM

Write a Qiskit function that takes a boolean function $f: \mathbb{F}_2^n \rightarrow \mathbb{F}_2^d$ and outputs a circuit, U , such that

$$U|x\rangle_n |0\rangle_d = |x\rangle_n |f(x)\rangle_d.$$

The construction may use any number of ancillas and any controlled multi-qubit gates, and the built-in version of the Quantum Fourier Transform (and its inverse), if needed.

Expectations:

- Documentation.
- Working Qiskit code for all n and demo for $n = 3$.

Note: This is a more open-ended problem, there are fairly straightforward, but inefficient ways to do (which are nevertheless have great education value), but you can also try to aim for something more SOTA.

MINI-PROJECT #9: ARBITRARY UNITARY SYNTHESIS USING MULTIPLEXED ROTATIONS (HARD)

Write a Qiskit function that takes an n -qubit unitary, U , as an input and outputs a quantum circuit that implements U , using only 1-, and 2-qubit gates, and multiplexed rotations. You *can*, but *don't have to* implement the multiplexed rotations, like we did in class.

Recommended approach: You can follow the ideas of this paper: [Unitary synthesis with fewer T gates](#). This construction uses a few smart, but not extremely abstract ideas from this reference.

Expectations:

- Documentation. This includes a proper README and a well-documented notebook.
- Working Qiskit code for all n and demo for $n = 3$.

MINI-PROJECT #10: MIXED QUANTUM STATE PREPARATION

Write a Qiskit function that takes a $2^n \times 2^n$ density matrix, $\rho \in \mathbb{C}^{2^n \times 2^n}$, and outputs a circuit, U (with measurements), that prepares ρ .

The construction may use any number of ancillas, arbitrary 1-qubit gates and multiplexed Z -rotations, (the number of controlled may be arbitrarily large).

Expectations:

- Documentation.
- Working Qiskit code for all n and demo for $n = 2$.

Ways to make it more challenging: Search for recent papers on arXiv.